

Name: _____

Class: _____

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Sound: Sampling, Sample Rate and Bit Depth

You already know how a picture becomes binary. Sound works the same way, and most of the words carry straight across.

Images	Sound
a pixel is one dot of colour	a sample is one measurement of the wave
colour depth is the bits per pixel	bit depth is the bits per sample
more pixels, more detail	more samples per second, more detail
size = $W \times H \times \text{colour depth}$	size = $\text{sample rate} \times \text{seconds} \times \text{bit depth}$

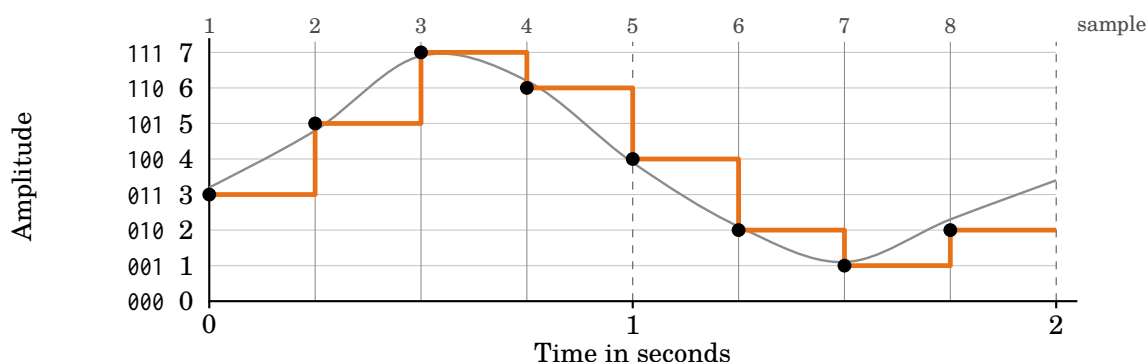
Bit depth is to sound what colour depth is to images.

LO1 • A sound is a wave, and a sample is one measurement of it

OCR 1.2.4 — representing sound · sampling

Sound is a vibration, and its height, the **amplitude**, changes smoothly all the time. A microphone turns it into an electrical signal, and an **analogue-to-digital converter (ADC)** measures that signal at regular intervals. Each measurement is a **sample**, stored as a binary number.

Here is 2 seconds of sound, measured 8 times. That is 4 samples every second, a **sample rate of 4 Hz**. Each solid line up the page is one sample, and the dashed lines mark each whole second. The grid has 8 levels, so each sample needs **3 bits**, because $2^3 = 8$. At each sample line, read the height of the wave and round it to the **nearest level**.



Sample	1	2	3	4	5	6	7	8
Height	3	5	7	6	4	2	1	2
Binary	011	101	111	110	100	010	001	010

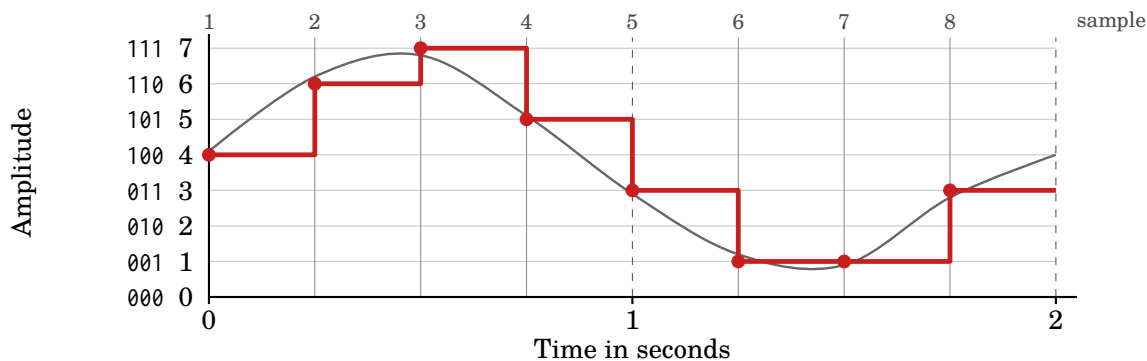
Those 24 bits *are* the sound file. They are the same eight numbers as page 11 of Booklet 2. Eight samples at 3 bits each is $8 \times 3 = 24$ bits, which is $24 \div 8 = 3$ bytes. The same answer comes from the rate: 4 samples a second, for 2 seconds, at 3 bits each, is $4 \times 2 \times 3 = 24$ bits.

The thick **orange line** is what the computer stores and plays back. Each sample is rounded to the nearest level and held flat until the next one. What the grey wave did between samples is lost, so the copy is close to the original and never identical to it.

OCR 1.2.4 — sampling a wave · rebuilding a wave

- (a) Sample this wave, which is also 4 Hz for 2 seconds. At each sample line, read the nearest level, then write it in denary and in binary.

[4]

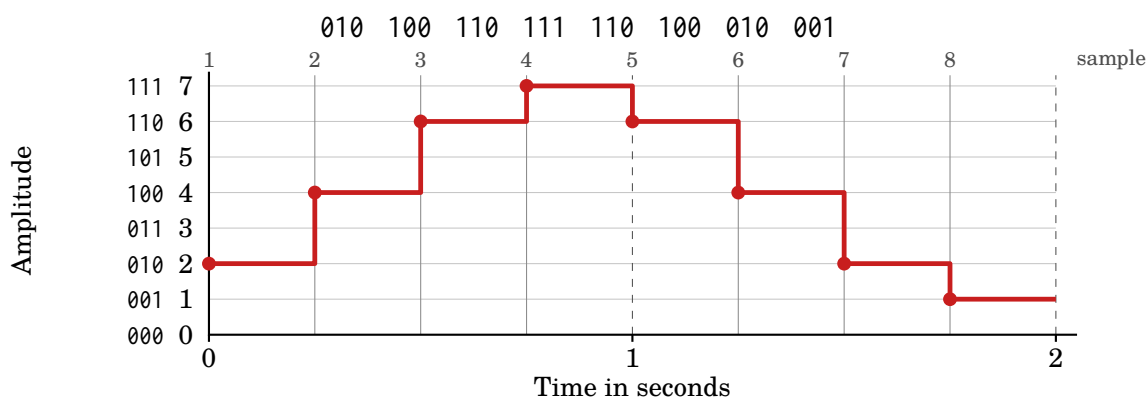


Sample	1	2	3	4	5	6	7	8
Height	4	6	7	5	3	1	1	3
Binary	100	110	111	101	011	001	001	011

Check: every binary code is exactly 3 digits long, and it matches the label at the left of its level.

- (b) Now the other way round. Here are 8 samples at a bit depth of 3. Mark each one as a dot on its sample line. Then draw the stored sound the way the orange line does: flat from each dot to the next sample line, then straight up or down to the next dot.

[4]



Check: your line climbs in steps to level 7 at 0.75 seconds, then steps back down, ending on level 1.

- (c) Both recordings above hold 8 samples at a bit depth of 3. State how many bits, and how many bytes, each one needs.

[2]

$8 \times 3 = 24$ **bits**, and $24 \div 8 = 3$ **bytes**.

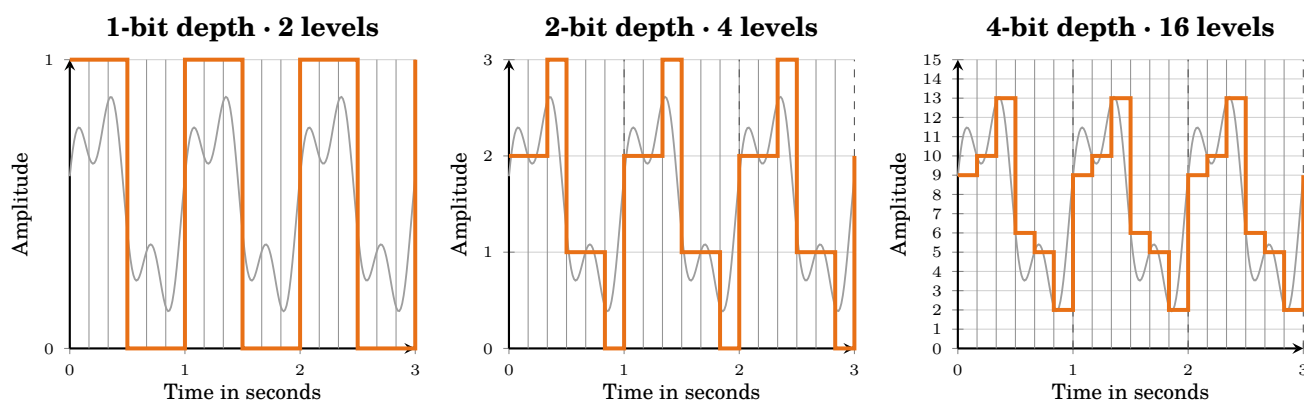
LO3 • Two dials: how precisely, and how often

OCR 1.2.4 — bit depth and sample rate, and their effect on quality

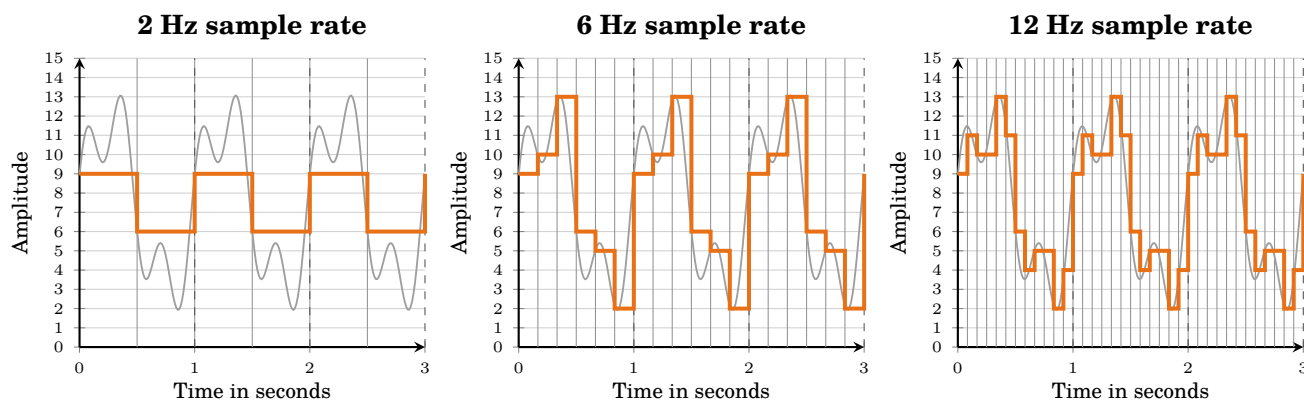
Bit depth is *how precisely* each sample is recorded: the number of bits per sample, sometimes called the **sample resolution**. n bits give 2^n different heights, the same doubling table as colour depth.

Bit depth	1 bit	2 bits	3 bits	8 bits	16 bits
Heights available	$2^1 = 2$	$2^2 = 4$	$2^3 = 8$	$2^8 = 256$	$2^{16} = 65,536$

Every graph below shows the same 3 seconds of sound, read the same way as page 1. The grey curve is the real sound and the orange line is what is stored. The levels up the side are the heights the bit depth allows, and the solid lines are the samples, so the lines between two dashed lines give the sample rate. The top row keeps 6 samples a second and changes the bit depth.



Sample rate is *how often* the wave is measured: the number of samples taken each second, measured in **hertz (Hz)**. 8,000 Hz means 8,000 samples every second. The bottom row keeps a 4-bit depth and changes the sample rate.



- (d) Look along the top row. Describe what a higher bit depth does to the stored sound, and say what it costs.

[3]

More bits give more heights to choose from, so each sample is rounded to a value closer to the real wave, and the orange line follows the grey curve more closely, so the sound is more accurate and better quality. It costs storage: every sample takes more bits, so the file is bigger.

- (e) Look along the bottom row. Describe what a higher sample rate does to the stored sound, and say what it costs.

[3]

More samples each second means the wave is measured more often, so less happens between two samples and the orange steps become narrower and trace the real wave more closely, so quality improves. It costs storage: more samples are stored every second, so the file is bigger.

LO4 • How big is the file?

OCR 1.2.4 — calculating sound file size

Count the samples, multiply by the bits in each one, then tidy the units. This is the formula exactly as Booklet 2 writes it on page 12.

$$\text{file size (bits)} = \text{sample rate} \times \text{duration} \times \text{bit depth}$$

Sample rate in Hz, duration in **seconds**, bit depth in bits. Divide by 8 for bytes, then by 1,000 for kilobytes.

Worked example 1, straight off a graph. The **6 Hz, 4-bit** graph on page 3 holds 3 seconds of sound. Count the solid lines in any one second and there are 6.

Size of sound file (bits) =		
Sample rate	6	samples per second, in Hz
Duration	3	length of the sound, in seconds
Bit depth	4	bits stored for each sample
$6 \times 3 \times 4$	=	72 bits = $72 \div 8 =$ 9 bytes

Worked example 2, a real recording. A doorbell chime is recorded at 10,000 Hz for 5 seconds with a bit depth of 8 bits.

Size of sound file (bits) =		
Sample rate	10,000	samples per second, in Hz
Duration	5	length of the sound, in seconds
Bit depth	8	bits stored for each sample
$10,000 \times 5 \times 8$	=	400,000 bits $\div 8 = 50,000$ bytes $\div 1,000 =$ 50 kB

- (f) A voice note is recorded at 8,000 Hz for 10 seconds with a bit depth of 16 bits. Calculate its file size in kilobytes. Show your working.

[3]

8,000 $\times$ 10 = 80,000 samples. 80,000 $\times$ 16 = 1,280,000 bits. 1,280,000 $\div$ 8 = 160,000 bytes = 160 kB.

- (g) A song is recorded at 44,100 Hz for 2 minutes with a bit depth of 16 bits. Construct an expression for its file size in bits. You do not need to work it out.

[2]

44,100 $\times$ 120 $\times$ 16. (One mark for the duration in seconds, 120, and one for the three values multiplied. The command word is construct an expression, so the marks are for the expression itself.)

Check: the duration you multiplied by is in seconds, so 2 minutes became 120.

- (h) The voice note in (f) is recorded again at 16,000 Hz, with everything else the same. State its new file size, and explain how you know without starting again.

[2]

320 kB. The sample rate has doubled, so there are twice as many samples, each the same size, so the file doubles.

- (i) Phone calls are usually sampled at 8,000 Hz. Music is usually sampled at 44,100 Hz. Explain why the two are chosen differently.

[3]

A phone call only has to carry speech clearly, and its data is sent live, so a low sample rate keeps the amount of data small and quick to send while the voice is still understandable. Music needs to sound as close to the original as possible, including high notes, so it uses a much higher sample rate and accepts a larger file. (Credit any answer that weighs quality against file size for each purpose.)

L05 • Hear it

Two free tools in the browser · no account needed · headphones for the second

Tool 1, see the samples and check your maths. Open the **Sound Sampling Simulator** at 101computing.net/i/sound-sampling-simulator.html. Its sample rate is counted in samples per second, so the numbers stay small enough to work out in your head.

- (j) For each row, work out the file size **first**, in bits and in bytes. Then set the three sliders and read the **Bits** and **File size** boxes.

[3]

Sample rate	Duration	Bit depth	My bits	My bytes	Matched?
12	4 s	4 bits	192	24	
20	5 s	8 bits	800	100	
40	10 s	8 bits	3200	400	

Check: the simulator's File size box shows the same number of bytes as your own column.

- (k) Press **Low quality sound**, then **Higher quality sound**. Describe how the digital wave changes, and what happens to the file size.

[2]

The higher quality setting has more samples and more levels, so the digital wave follows the original much more closely. The file size is much larger.

Tool 2, hear it. Put on headphones and open the **Bit Depth and Sample Rate Simulator** at cmuse.org/bit-depth-and-sample-rate-simulator. Choose the **sine** wave at **440 Hz**. The **O** button plays the original and **C** plays the converted copy, so you can flip between them.

- (l) Leave the sample clock where it is. Lower the **bit depth** one step at a time from 16, flipping between O and C each time. Record the lowest bit depth at which C still sounds the same as O to you.

Lowest bit depth that still sounds right: _____ bits **Answers vary. Hiss usually becomes audible somewhere around 6 to 8 bits.**

- (m) Set the bit depth back to 16. Now lower the **sample clock** a step at a time and listen. Describe what you hear, and explain it using what you know about sample rate.

[2]

The sound becomes rougher and eventually changes pitch or gains extra tones. Fewer samples each second means the wave is measured less often, so the stored copy is a poorer match for the original and plays back less accurately.